

K-means clustering

- the number of clusters K is given
- idea: a "good" clustering is one for which the **within-cluster variation** is as small as possible
- each cluster C_k is characterized by its **centroid** $\bar{x}_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$
- tends to minimize the overall distance between each point and its cluster's centroid:

$$\min_{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K} W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K) = \sum_{k=1, \dots, K} \sum_{x_j \in C_k} d(x_j, \bar{x}_k)$$

- the algorithm is an alternation of two main steps
 - a assign the points to the cluster with the closest centroid
 - b determine the centroids of each cluster
- the algorithm stops when the two steps do not change the assignment (or do not improve $W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K)$) anymore



K-means clustering scheme $\rightarrow$ SUBOPTIMAL SOLUTIONS

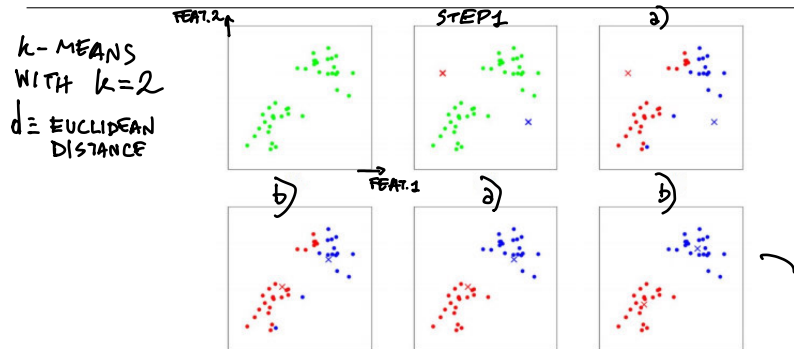
0 select a distance d and a number K

1 (initialization) randomly determine K centroids $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K \in \mathbb{R}^n$

2 iterate until the cluster assignment stop changing

a (cluster assignments) assign each observation to the closest centroid

b (centroids update) update the centroids of each cluster as $\bar{x}_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$



some considerations on K-means

- sensitive to anomalous points/outliers
- points can move from a cluster to another
- converges to a sub-optimal solution $\Rightarrow$ the final solution depends on the random initialization
- if points have not a strong "clustered" structure or K is "wrong" the algorithm is more sensitive to initialization
- good practice: multiple runs of K-means for different initialization and different K and select the best solution (in terms of $W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K)$)

↓
OPEN QUESTION

$$\uparrow K \Rightarrow \downarrow W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K)$$



CLUSTERINGS WITH
DIFFERENT K CANNOT
BE COMPARED IN TERMS OF
 $W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K)$



how to verify the quality of a clustering

have we obtained relevant structures? is K right?

we can perform computational assessments → ENOUGH CPU-TIME
OR COMPUTING POWER

- by **perturbating** data and verifying if the clustering is stable

- ▶ add noise to data
- ▶ delete points (resample without replacement)
- ▶ bootstrapping (resample with replacement)

} WE COMPARE
THE ORIGINAL
CLUSTER WITH
THOSE OBTAINED
AFTER THE
PERTURBATIONS

- use the **silhouette** measure and plot...

⇒ IF THEY ARE SIMILAR
THE ORIGINAL
CLUSTERING WAS
GOOD

the silhouette of clusters

OBSERVATION

suppose point i belongs to C_i

C_i IS THE CLUSTER OF
OBSERVATION i

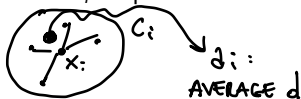


- mean distance from x_i to points of cluster C

$$d_{iC} = \frac{1}{|C|} \sum_{x_j \in C} d(x_i, x_j)$$

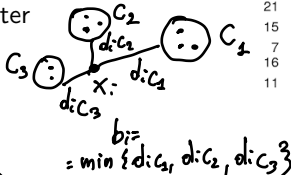
- mean distance from x_i to points of its OWN cluster

$$a_i = d_{iC_i}$$



- minimum mean distance from x_i to points of another cluster

$$b_i = \min_{C \neq C_i} d_{iC}$$



- silhouette of point i

$$SIL_i = \frac{b_i - a_i}{\max\{a_i, b_i\}}$$

$$SIL_i \leq 1$$

- overall silhouette

$$SIL = \frac{1}{m} \sum_{i=1, m} SIL_i$$

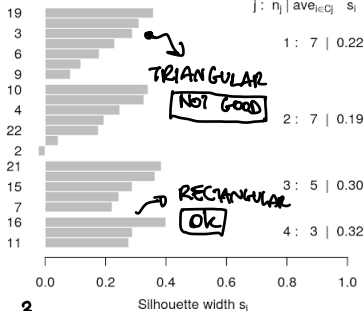
a_i TENDS TO
BE SMALLER
THAN b_i

Silhouette plot of (x = util_cl_m, dist = d)

n = 22

4 clusters C_j

$j: n_j | \text{ave}_{i \in C_j} s_i$



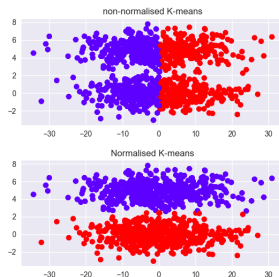
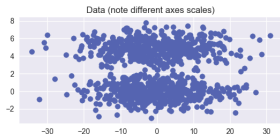
Average silhouette width : 0.24

$\uparrow SIL_i \Rightarrow b_i \gg a_i$
 $\Rightarrow$ GOOD CLUSTERING



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the issue of scaling data



MEAN

SPREAD

features with different locations or scales may influence the distance of two points (e.g. age and salary) and bias clustering results (the same holds for PCA!)

ALWAYS

MEAN = 0

often it is better to **center** and/or **scale** features

- Z-score scaler: $\frac{x - \bar{x}}{\sigma} \Rightarrow \text{mean} = 0, \text{st.dev.} = 1$
- unit scaler: $\frac{x - \min_x}{\max_x - \min_x} \Rightarrow \text{values in the interval } [0, 1]$

exercise 3.1

- 'An Introduction to Statistical Learning', Section 10.3